

How Many Futures Are Enough?

Learning Redundancy-Aware Effective Multimodality for
End-to-End Autonomous Driving with Action-
Conditioned Predictive Representations

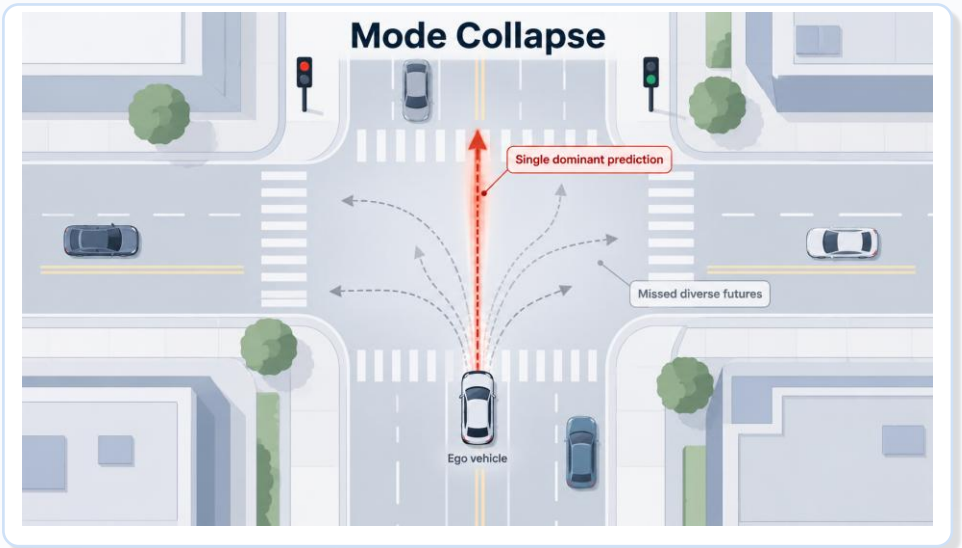
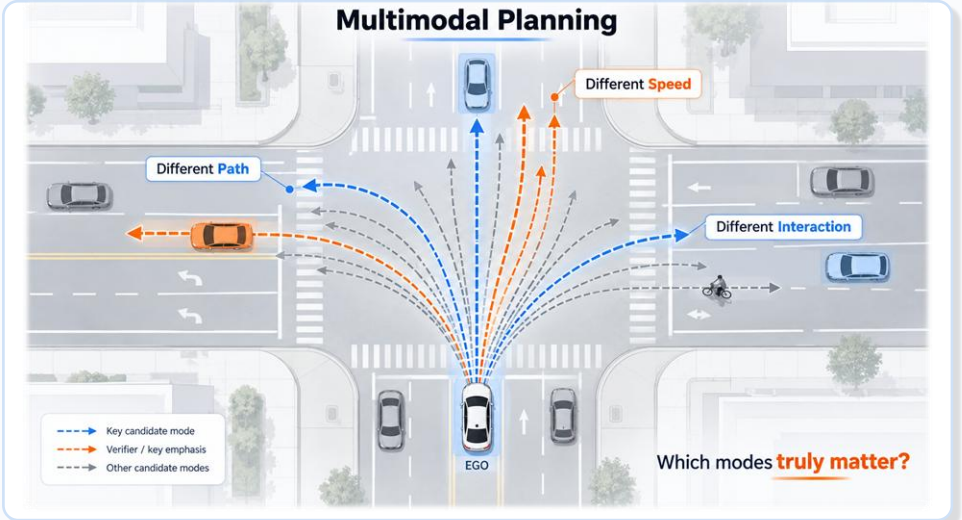
作者

Logo

From Trajectory Diversity to Decision Diversity

Multimodal decision-making

- Speed control**
Different speed profiles
- Path choice**
Multiple feasible routes
- Interaction**
Yield / pass / merge



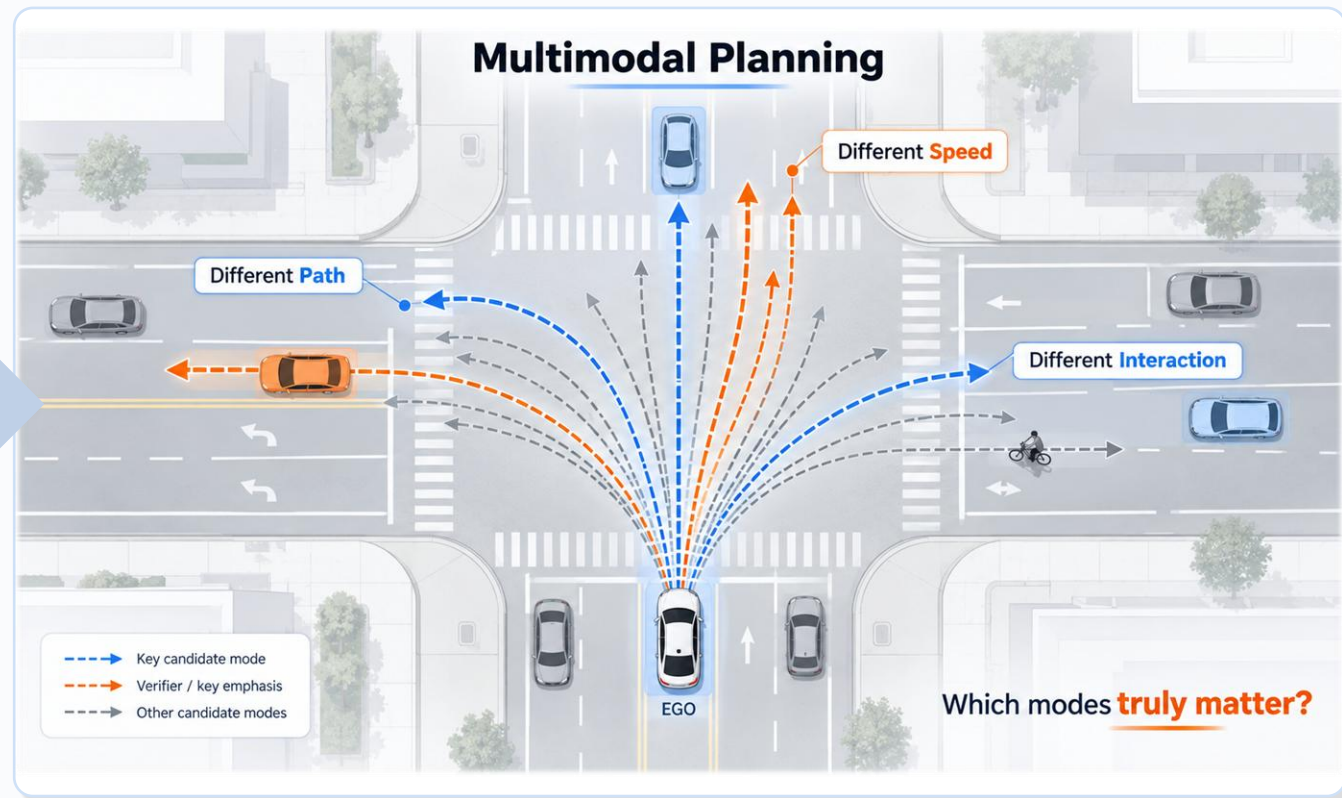
BACKGROUND

Representative Multimodal Planning Paradigms

Diffusion Model^[1]

Flow Matching^[2]

Proposal-based Planning^[3-4]



[1] B. Liao et al., "DiffusionDrive: Truncated Diffusion Model for End-to-End Autonomous Driving," *CVPR*, 2025.

[2] Z. Xing et al., "GoalFlow: Goal-Driven Flow Matching for Multimodal Trajectories Generation in End-to-End Autonomous Driving," *CVPR*, 2025.

[3] K. Guo et al., "iPad: Iterative Proposal-Centric End-to-End Autonomous Driving," *IEEE RA-L*, 2026.

[4] L. Wang et al., "Drive-JEPA: Video JEPA Meets Multimodal Trajectory Distillation for End-to-End Driving," *arXiv:2601.22032*, 2026.

Fixed Candidate Count



K = 32

Predefined output budget



Same predefined K

Used across different planning steps and scenarios.



Different decision needs

Different scenes contain different effective decision branches.



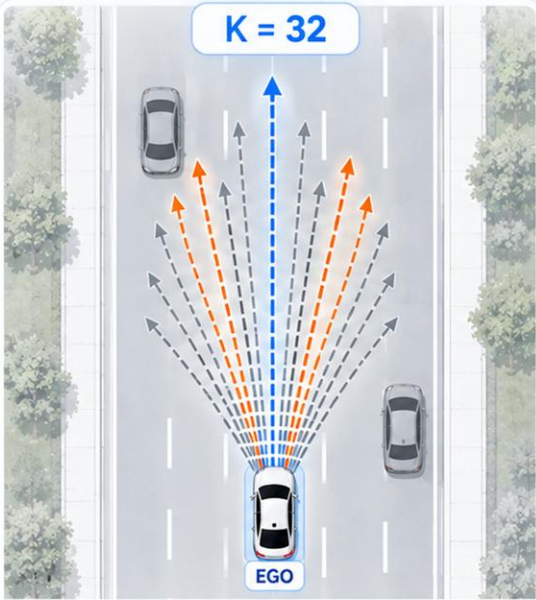
Possible redundancy

Fixed candidate count may include unnecessary or low-utility candidates, reducing efficiency.

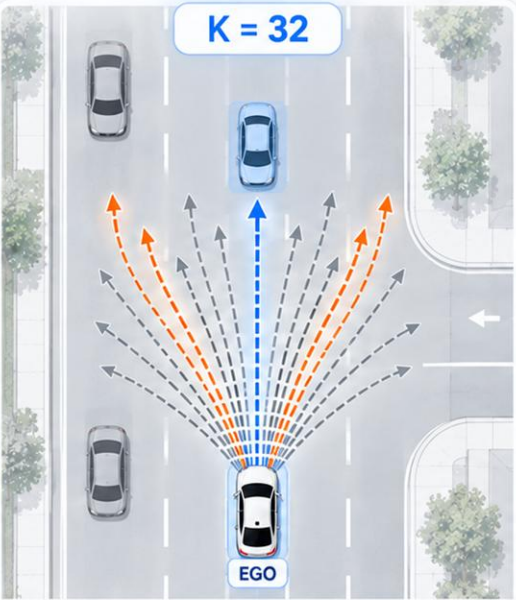


**Same K,
Different Decision Needs**

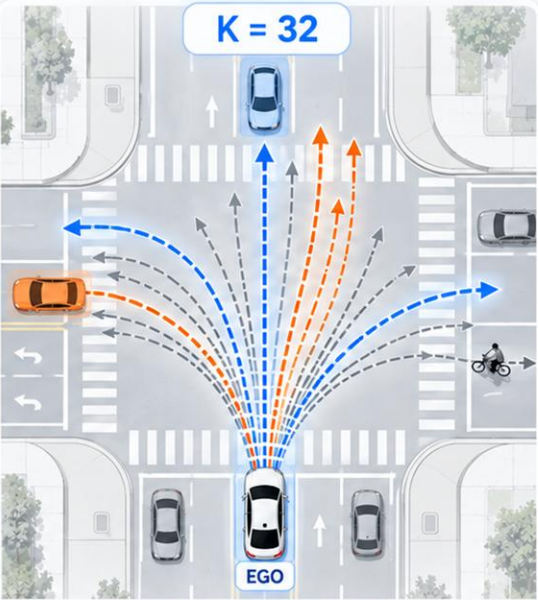
Fixed Candidate Count



Scene 1: Straight Open Road
Maintain lane, speed, or change lane



Scene 2: Car-Following / Lane Interaction
Keep distance, change lane, or overtake

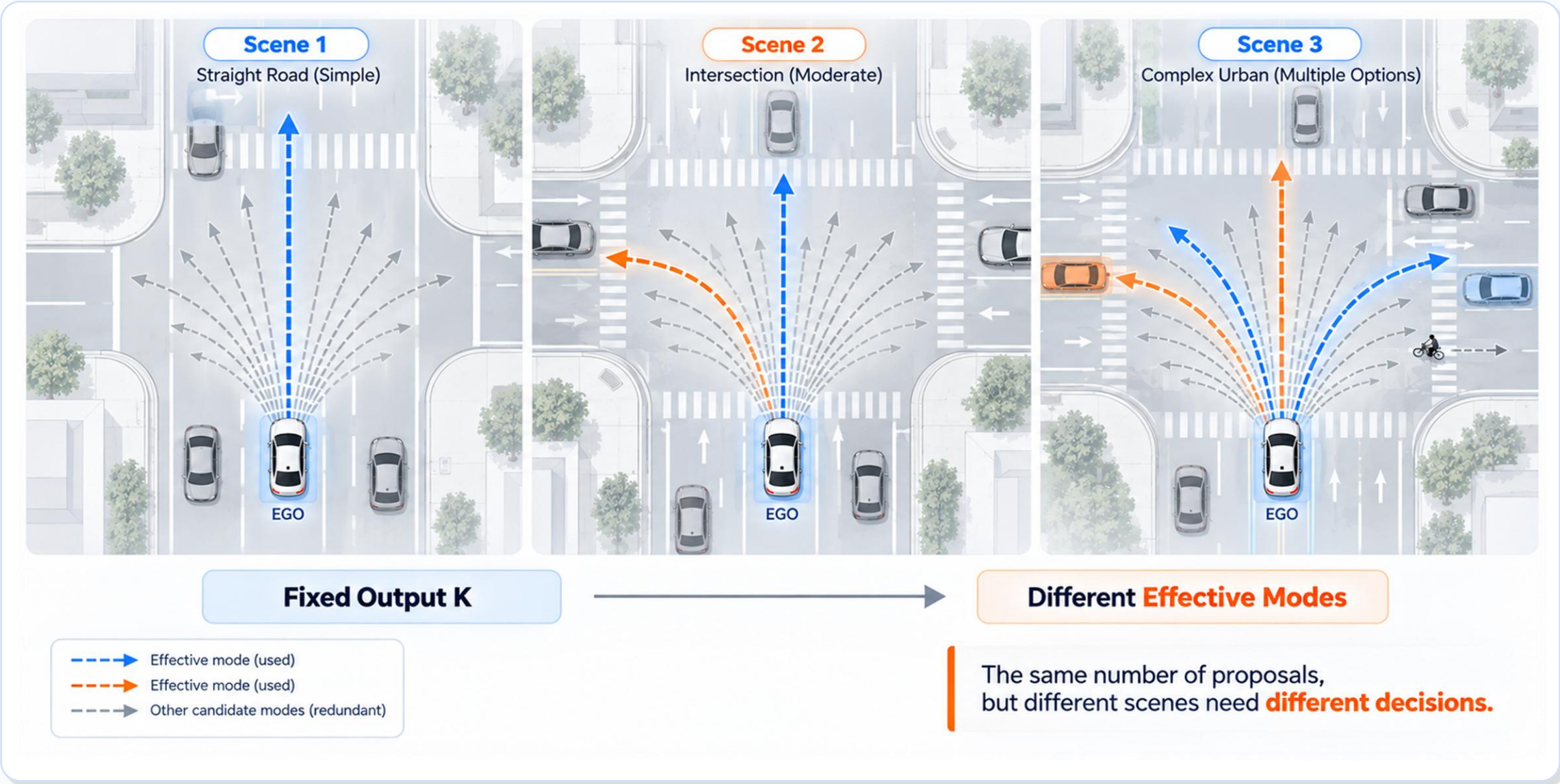


Scene 3: Complex Urban Intersection
Go straight, turn, yield, or respond to multiple agents

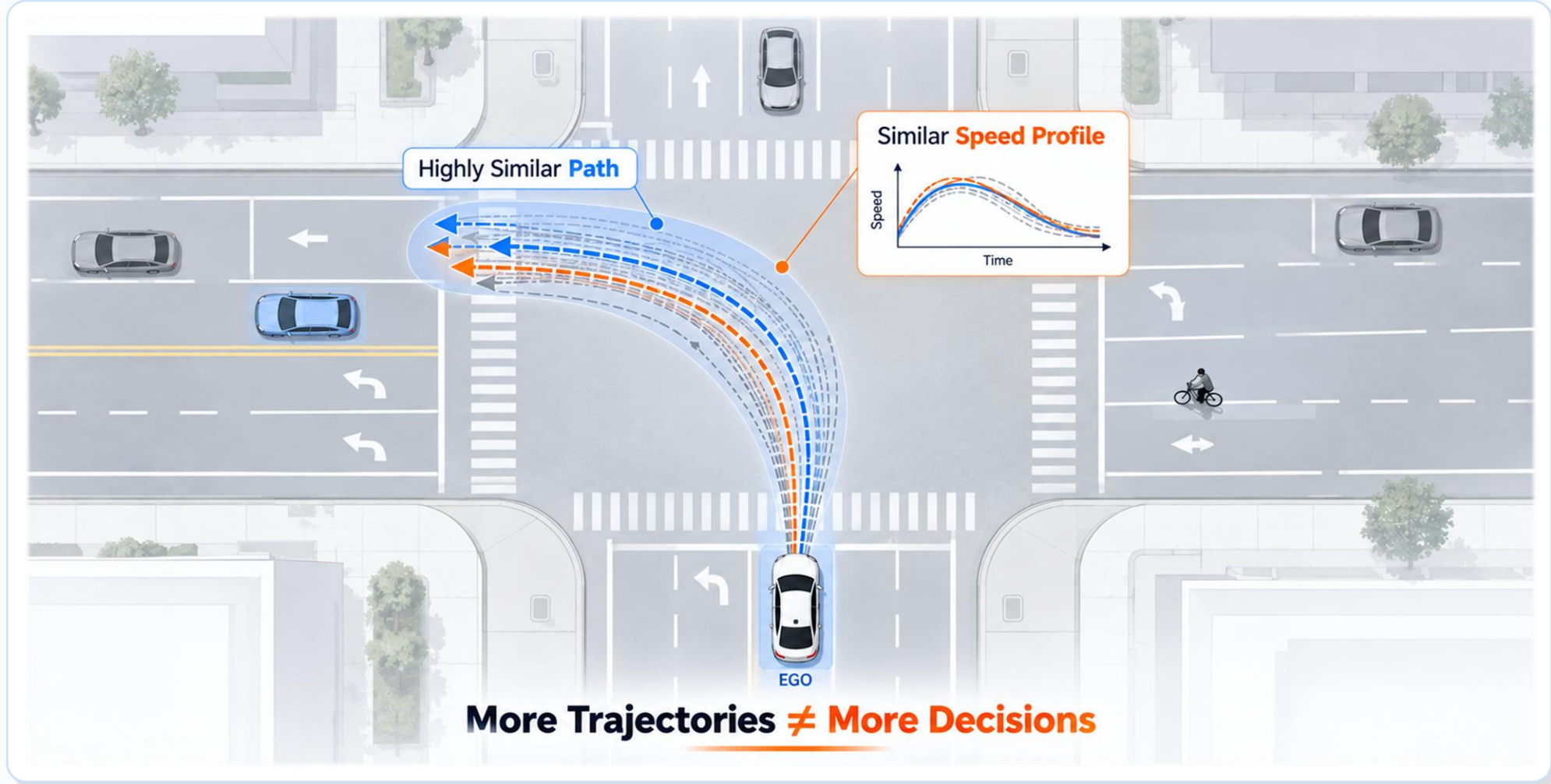
- Key candidate mode
- Verifier / key emphasis
- Other candidate modes

Same K, Different Decision Needs

Scene-level Redundancy

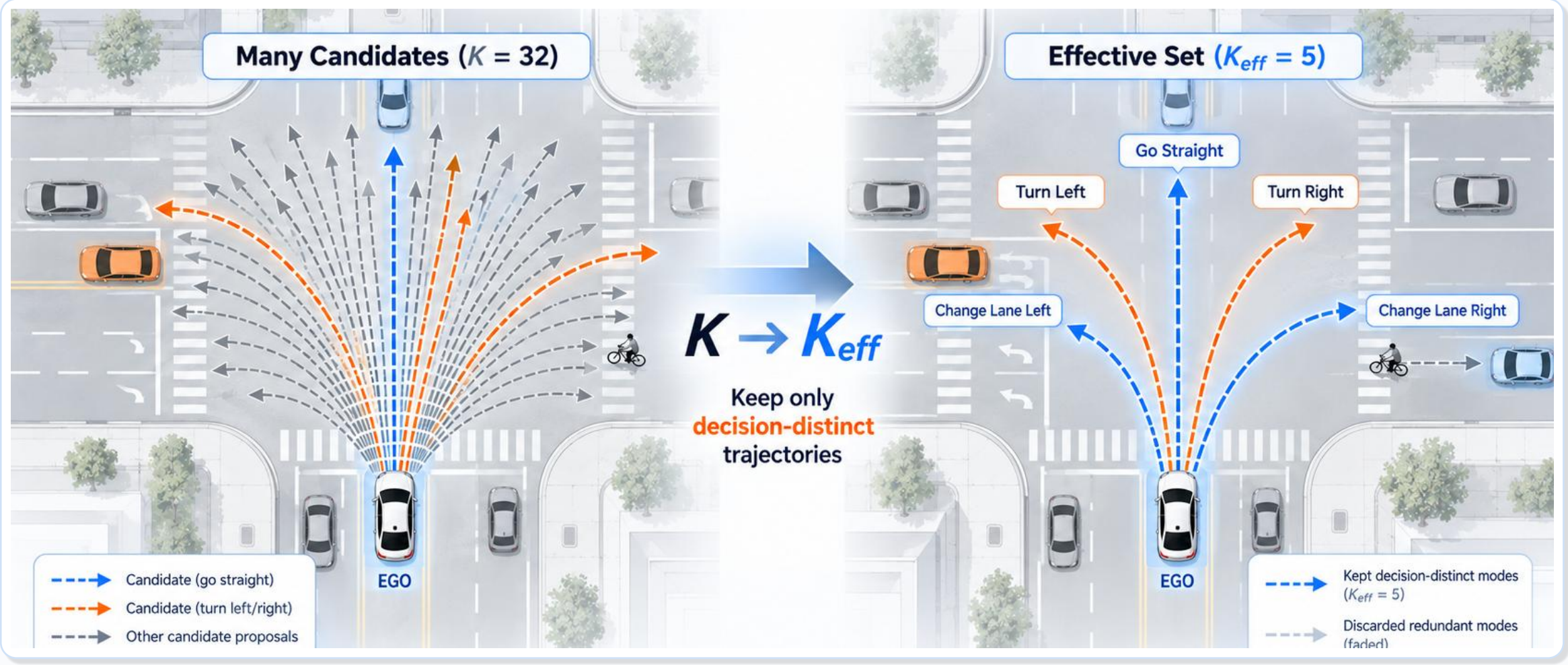


Strategy-level Redundancy



Dense left-turn trajectories may still express the same left-turn decision.

Effective Multimodality



$K \rightarrow K_{eff}$: keep only candidates that lead to different decision consequences.

Why Remove Redundancy?

Redundant Attempts Consume Capacity

When forced to provide many methods, the student spends extra time and mental effort on repetitive solutions.

Solve for x:
 $2x + 4 = 12$



Method 1 (Direct, clean)
 $2x + 4 = 12$
 $2x = 8$
 $x = 4$ ✓ Correct

Method 2 (Roundabout)
 $2x + 4 = 12$
 $2x = 12 - 4 + 0$
 $2x = 8$
 $x = 4$ Redundant

⋮

Method 3 (Overly complex)
 $2(x + 2) = 12$
 $x + 2 = 6$
 $x = 4$ Redundant

⋮

Method N (Another way)
 $x = \frac{12 - 4}{2}$
 $x = 4$ Redundant

Student Resources Get Depleted

⌚ Time 

🧠 Attention 

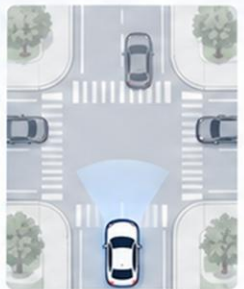
📁 Capacity 

Same Principle

From Redundant Trajectories to Greater Capabilities

A driving model may generate many similar candidate trajectories. Removing redundant ones frees capacity for higher-level reasoning.

1. Scene Input
Perceive the scene



2. Many Candidate Trajectories
Redundant and similar candidates



3. Scoring & Selection

	0.92
	0.78
	0.21
	0.18
	0.12
⋮	

Selected Trajectory



Remove redundant candidates
Less wasted computation, more effective learning.

Freed Capacity

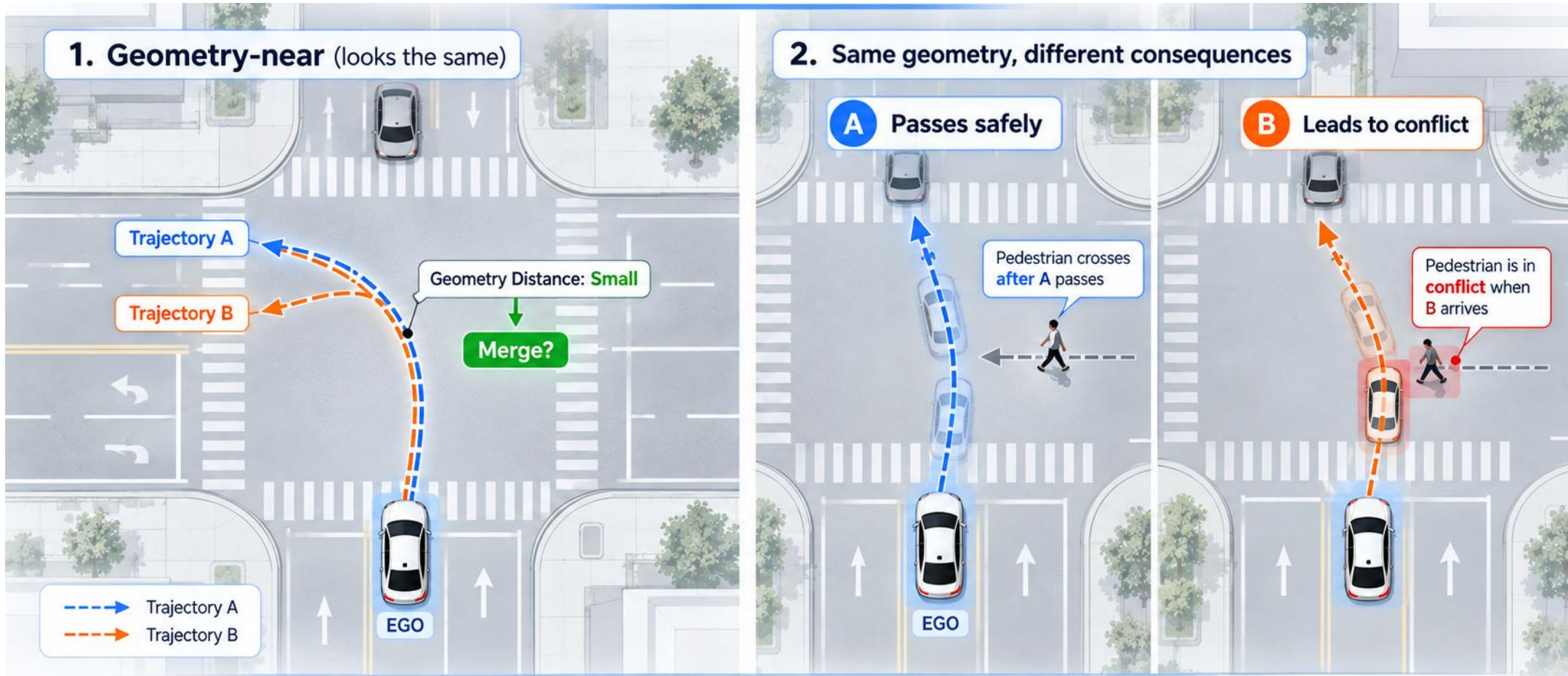


Use Capacity Where It Matters

- Causal Modeling**
Understand cause and effect in complex interactions
- Scene Representation**
Learn richer, structured world models
- Long-horizon Risk**
Reason about rare events and long-term consequences

Less Redundancy → More Capacity → More Capable and Safer Models

Why Geometry Alone Is Not Enough

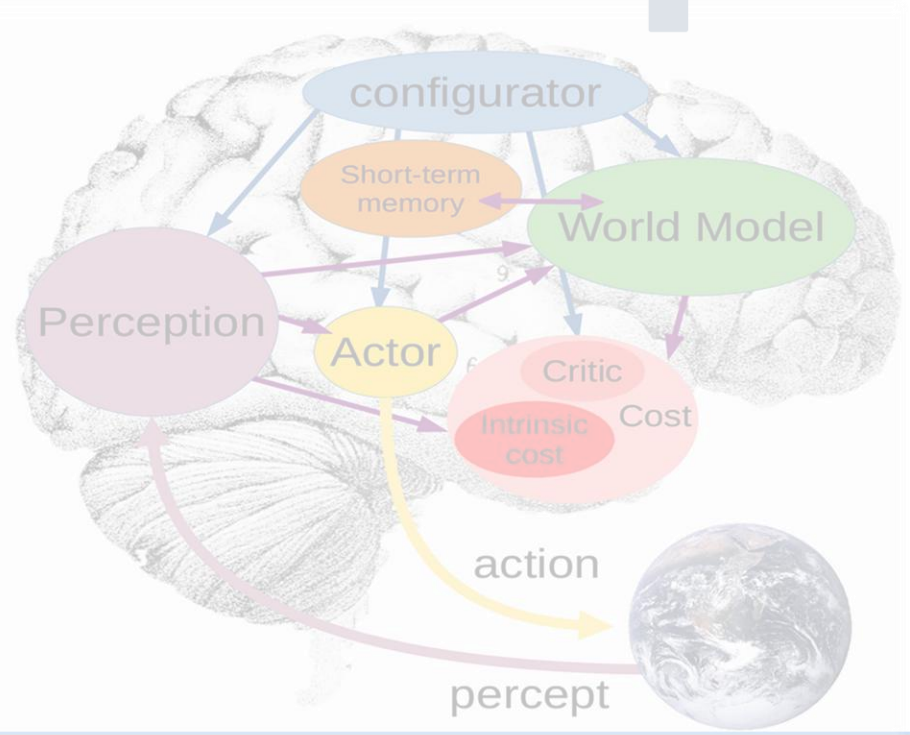
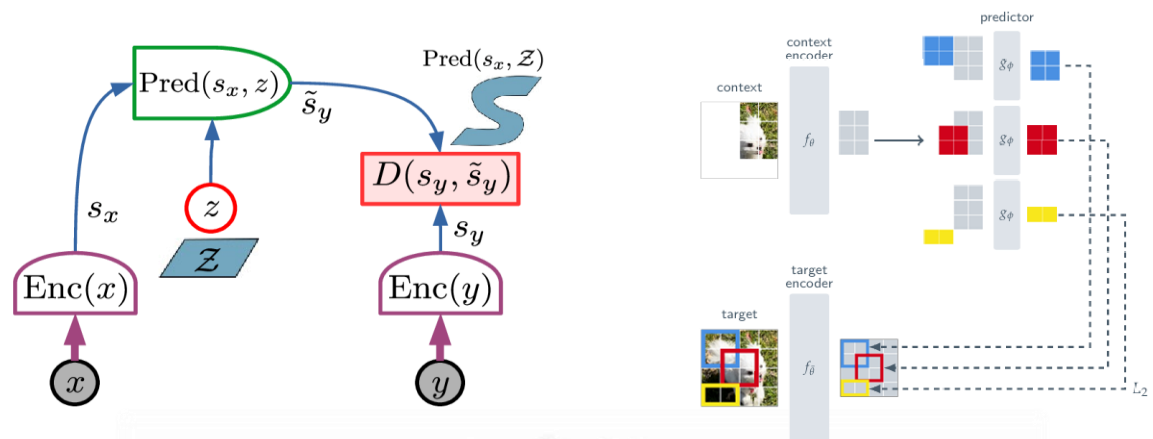


Geometry-near $\neq$ Consequence-equivalent

Trajectories can be geometrically similar, but lead to different outcomes due to **timing and interaction** with other agents.

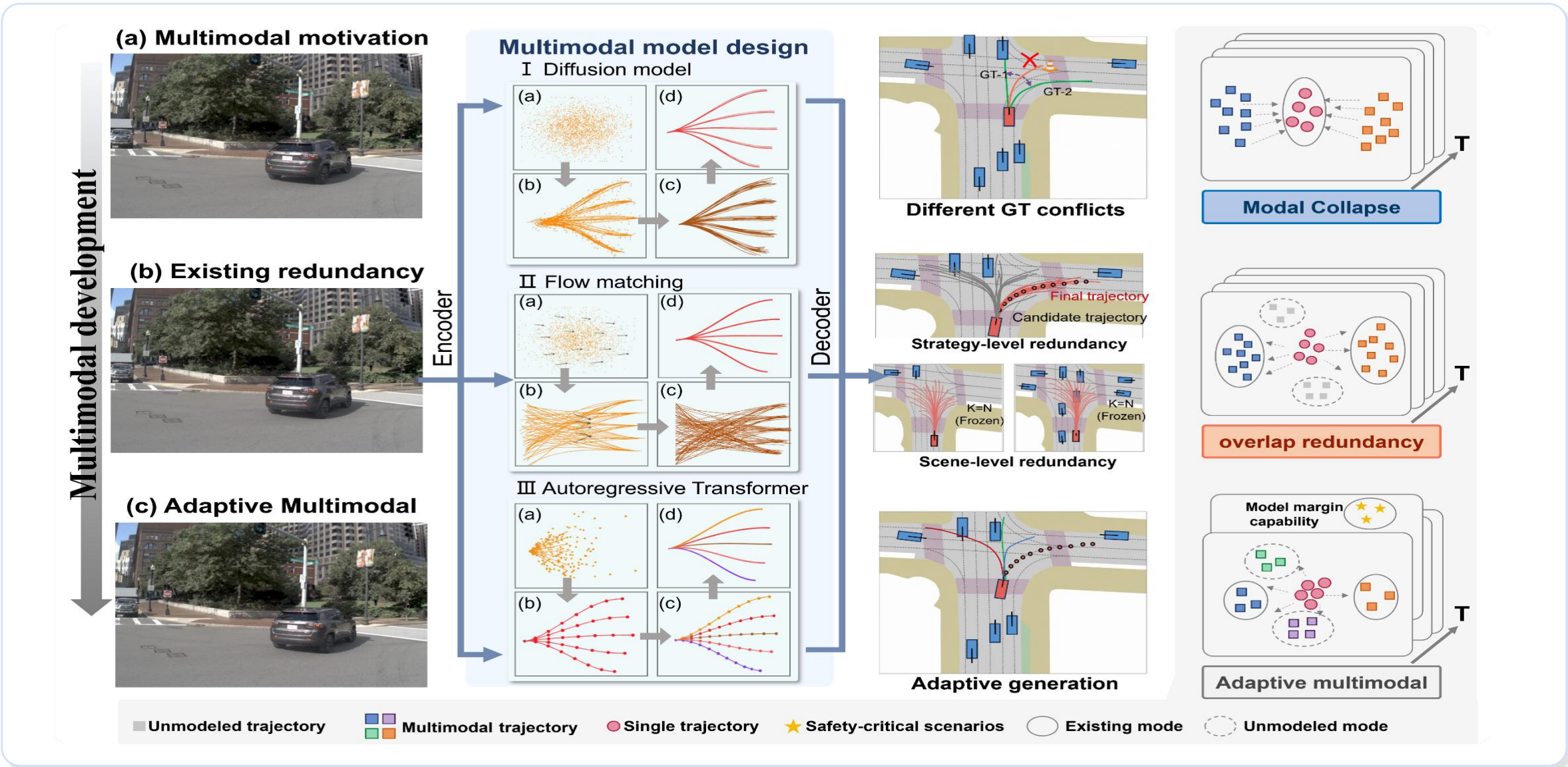
BACKGROUND

JEPA (Joint-Embedding Predictive Architecture)



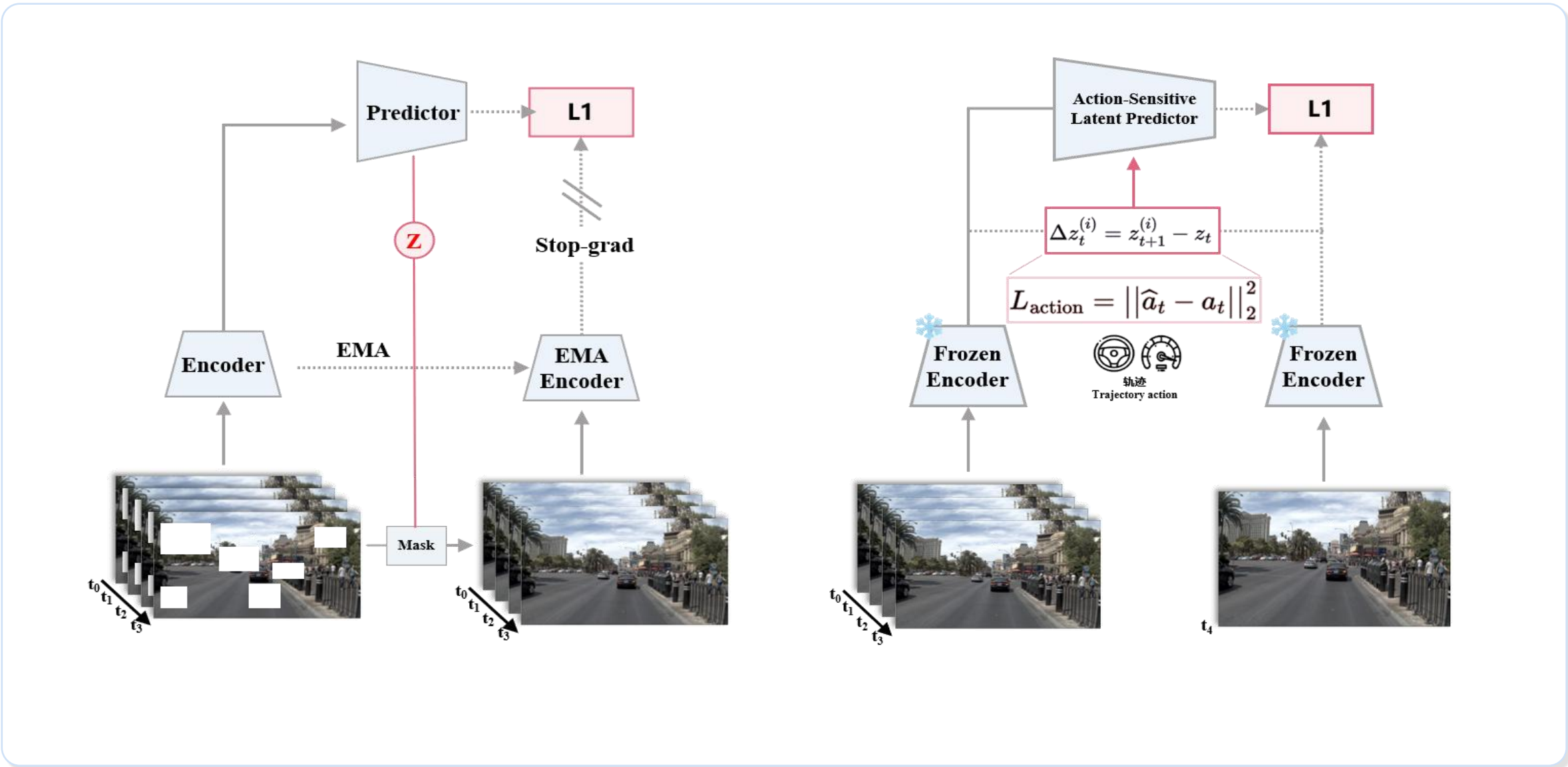
Yann LeCun: Pioneer of CNNs, Turing Award Laureate and One of the “Godfathers of AI”

AMC-Drive (Action-conditioned Multimodal Consolidation for Driving)

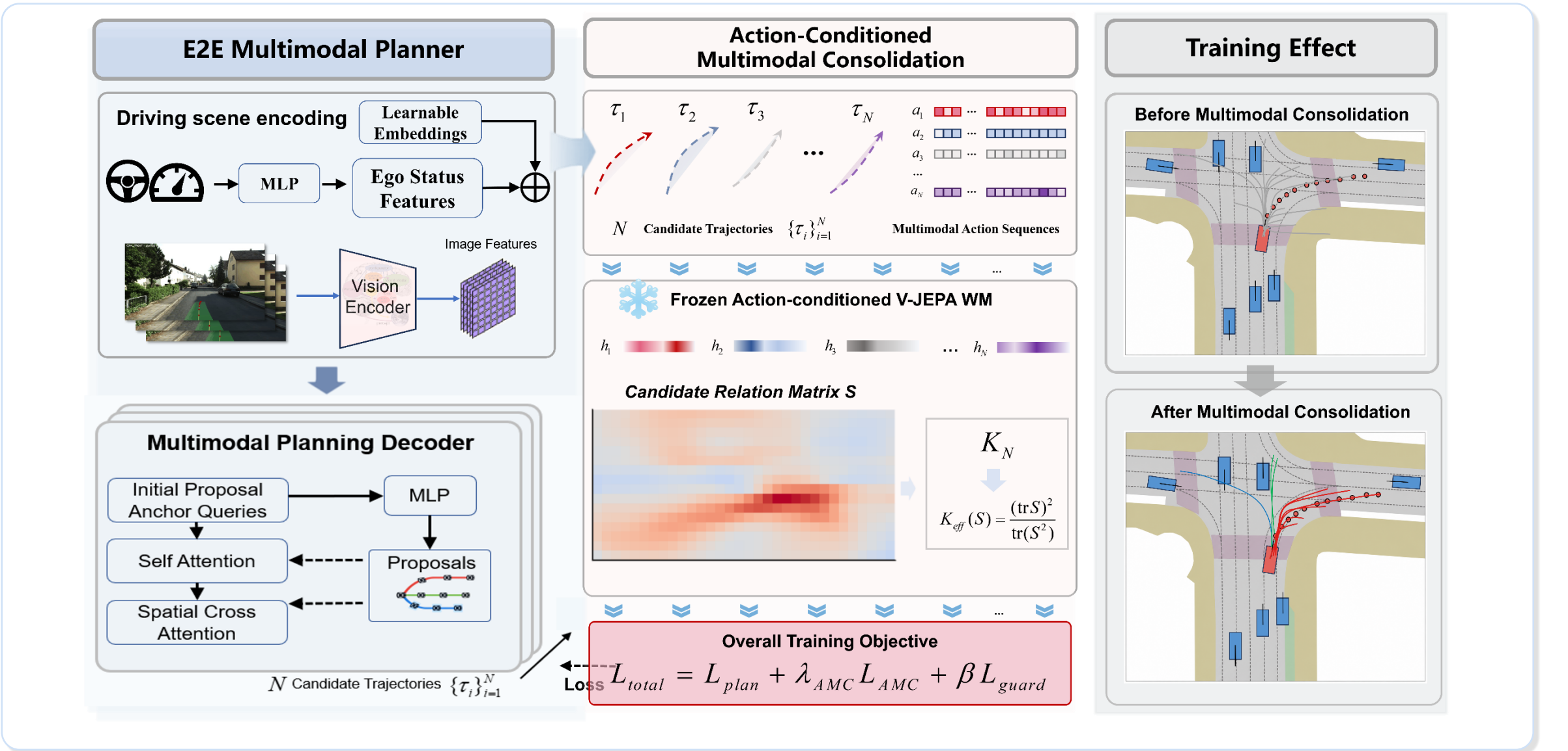


METHOD

AMC-JEPA (Action-conditioned Multimodal Consolidation JEPA)

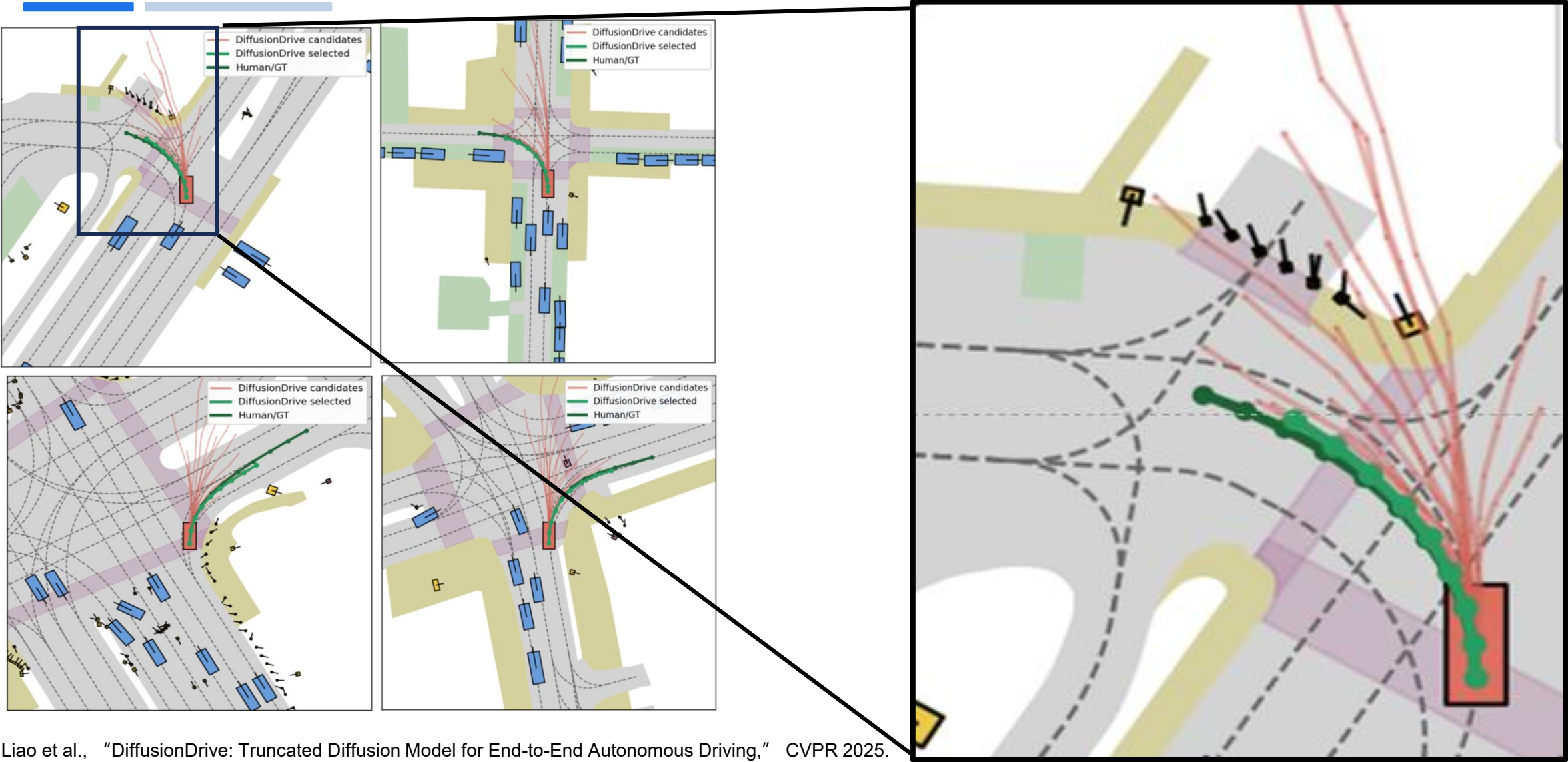


AMC-Drive (Action-conditioned Multimodal Consolidation for Driving)



METHOD

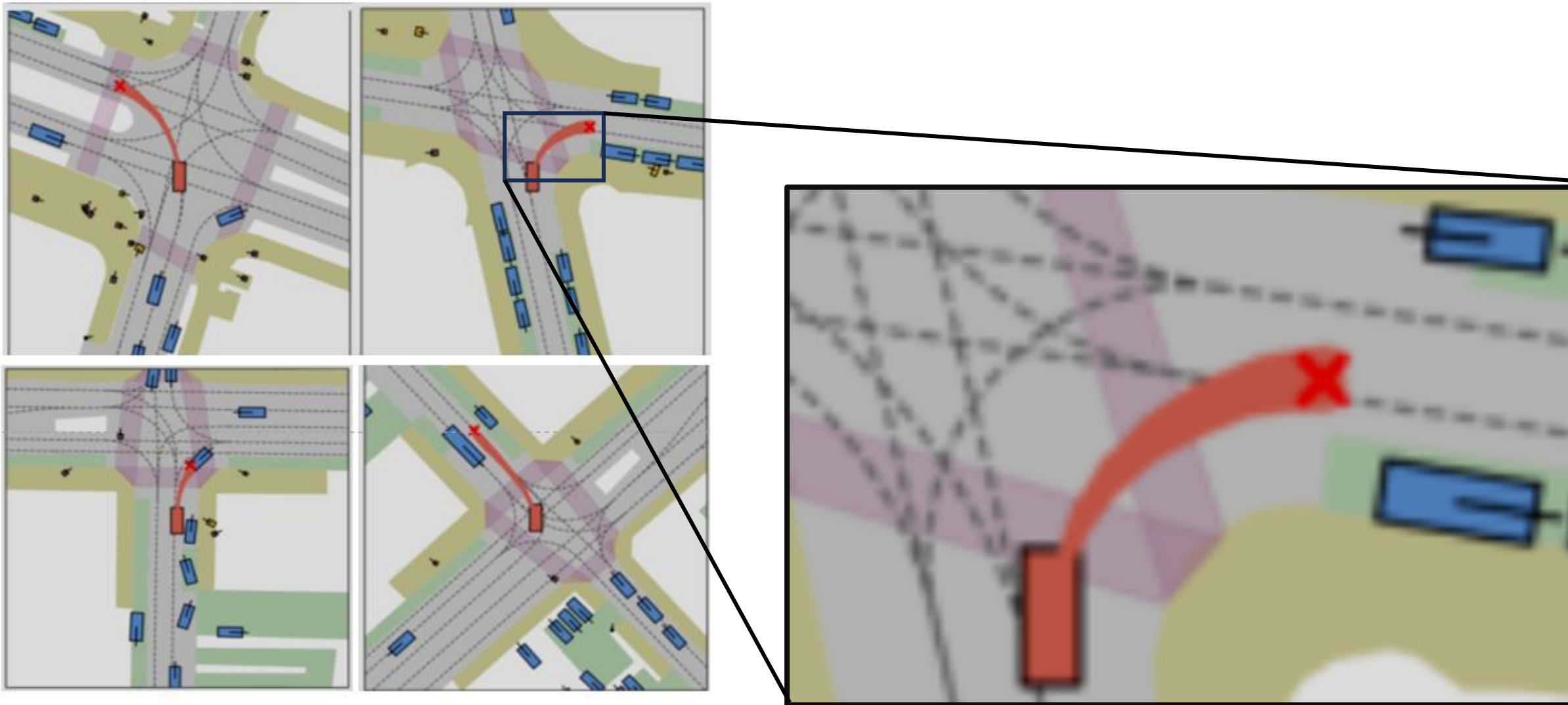
Diffusion Drive (Diffusion Model算法代表，多模态候选轨迹混乱无序)



[5] Liao et al., "DiffusionDrive: Truncated Diffusion Model for End-to-End Autonomous Driving," CVPR 2025.
[6] Y. Li et al., "ReCogDrive: A Reinforced Cognitive Framework for End-to-End Autonomous Driving," ICLR, 2026.
[7] L. Wang et al., "Drive-JEPA: Video JEPA Meets Multimodal Trajectory Distillation for End-to-End Driving," arXiv preprint arXiv:2601.22032, 2026

METHOD

ReCogDrive (FlowMatching 算法代表，坍塌模态并存在内部冗余)



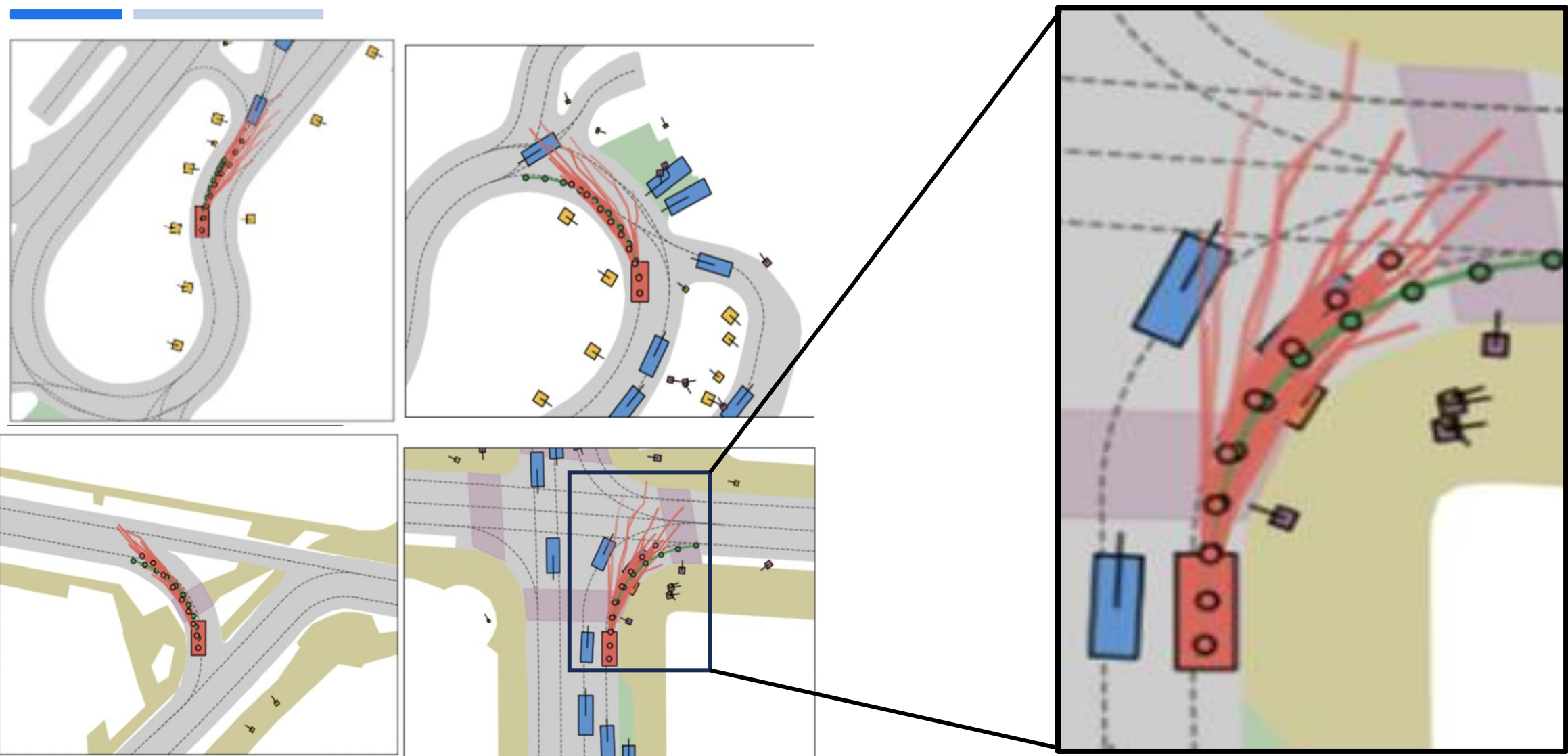
[5] Liao et al., "DiffusionDrive: Truncated Diffusion Model for End-to-End Autonomous Driving," CVPR 2025.

[6] Y. Li et al., "ReCogDrive: A Reinforced Cognitive Framework for End-to-End Autonomous Driving," ICLR, 2026.

[7] L. Wang et al., "Drive-JEPA: Video JEPA Meets Multimodal Trajectory Distillation for End-to-End Driving," **arXiv preprint arXiv:2601.22032**, 2026

METHOD

Drive-JEPA (Proposal base planner算法代表, 多模态候选轨迹混乱无序)



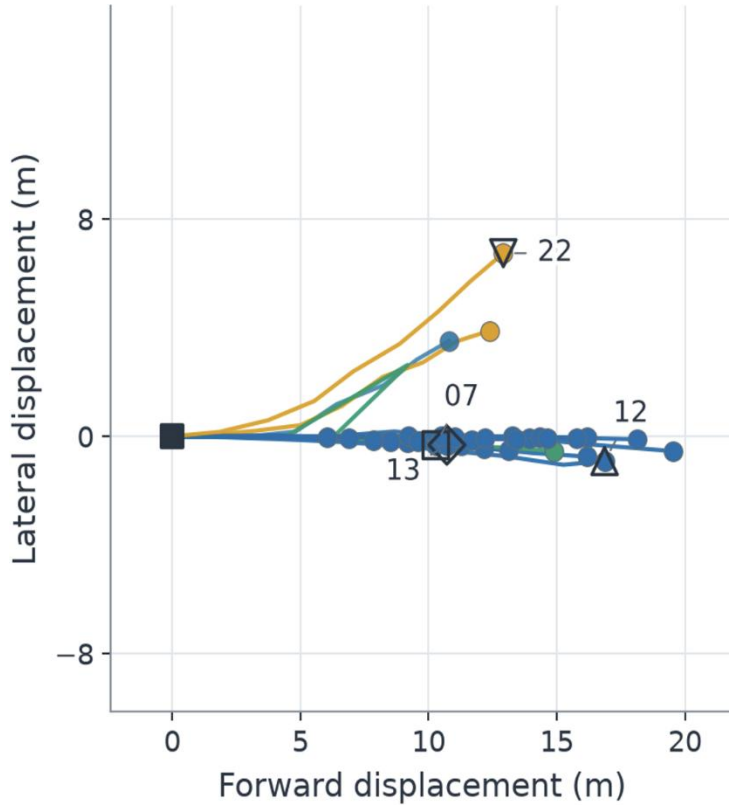
[5] Liao et al., "DiffusionDrive: Truncated Diffusion Model for End-to-End Autonomous Driving," CVPR 2025.
[6] Y. Li et al., "ReCogDrive: A Reinforced Cognitive Framework for End-to-End Autonomous Driving," ICLR, 2026.
[7] L. Wang et al., "Drive-JEPA: Video JEPA Meets Multimodal Trajectory Distillation for End-to-End Driving," **arXiv preprint arXiv:2601.22032**, 2026

AMC-Drive (Action-conditioned Multimodal Consolidation for Driving)

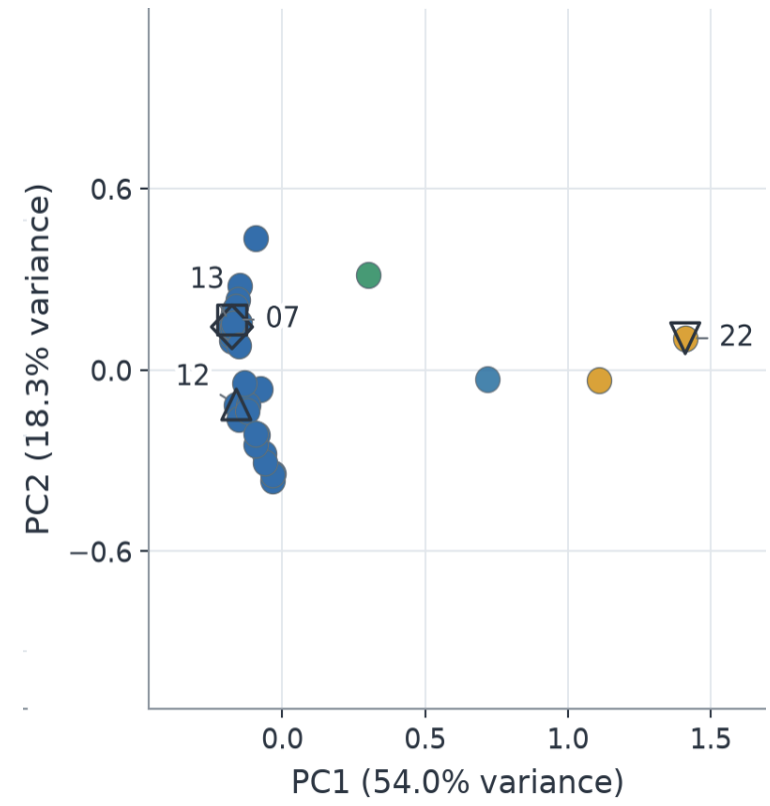
scene 026 · 0a96f9c66d895318



(a) NAVSIM front camera



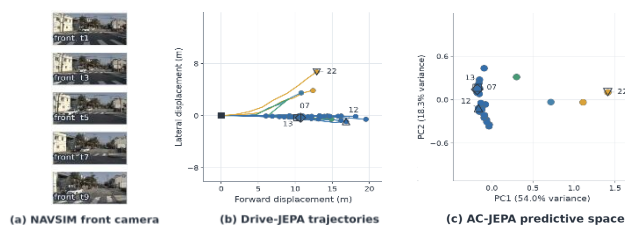
(b) Drive-JEPA trajectories



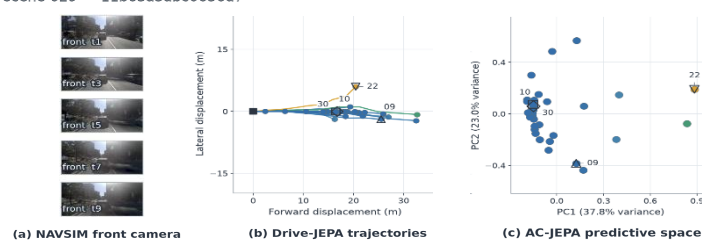
(c) AC-JEPA predictive space

AMC-Drive (Action-conditioned Multimodal Consolidation for Driving)

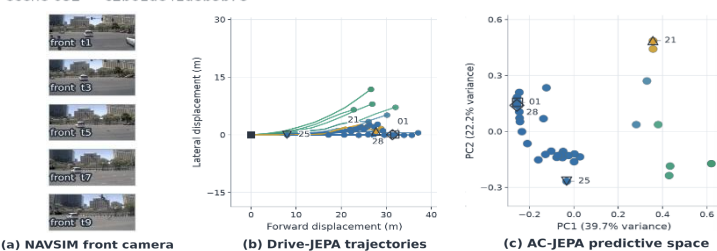
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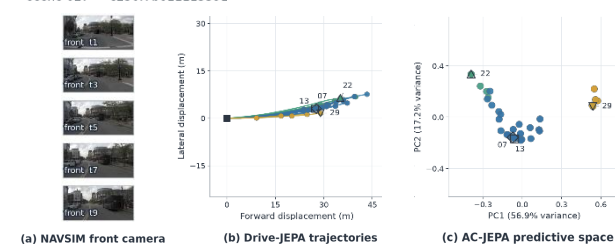
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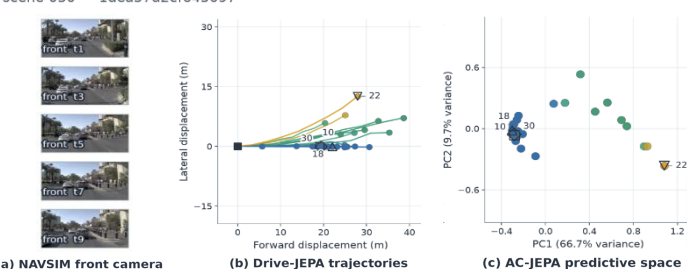
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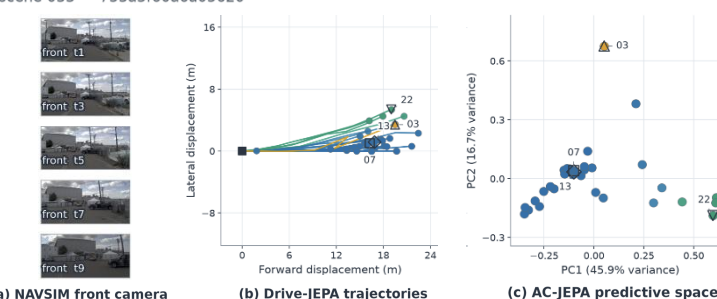
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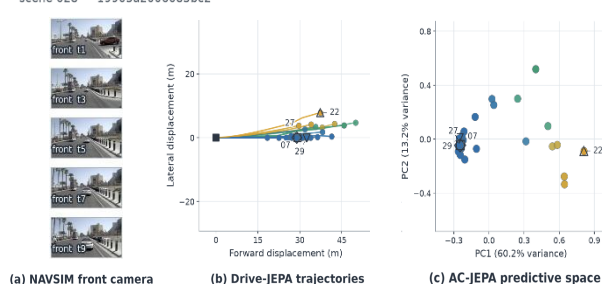
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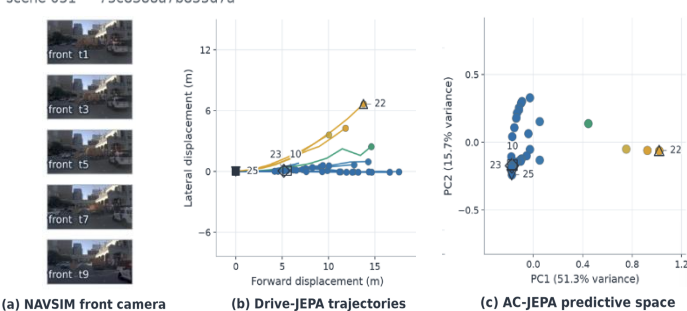
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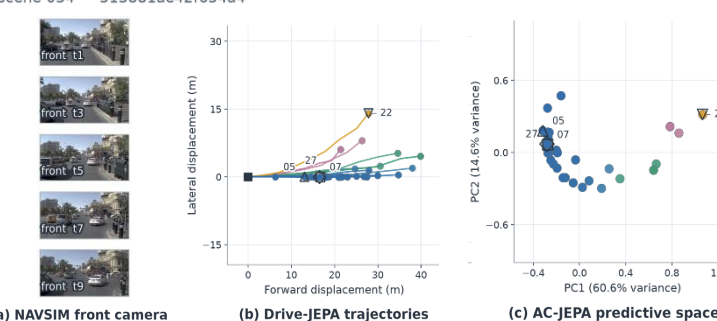
scene 028 · 19905a2006085bc2



scene 031 · 73c8388a7b855d7a



scene 034 · 513881ae42f654a4



KEY RESULTS

State-of-the-art performance on NAVSIM v1



JEPA Rank	Method	JEPA Type	NAVSIM-v1 PDMS
1	AMC-Drive	AC-JEPA + multimodal planning	93.9
2	Drive-HWM	V-JEPA slow world model + fast WAM	93.8
3	DA-WAM	V-JEPA 2.1 decision-aligned future latent	93.7
4	Drive-JEPA	V-JEPA + multimodal trajectory distillation	93.3
5	WA-JEPA	V-JEPA-native world-action modeling	91.8
6	Auto-JEPA	trajectory-latent joint embedding prediction	91.3

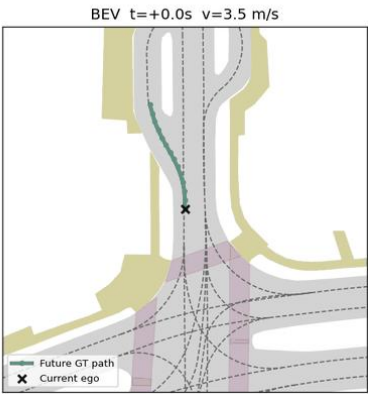


rank	id	avg_seeds_score	std_seeds_score	best_seed_score
1	EABOT.AI&NJU	95.4965	nan	95.4965
2	CooWAIM	95.4281	nan	95.4281
3	minfishmovemountain	95.3688	nan	95.3688
4	DriveZero	95.3087	nan	95.3087
5	bnu_ai	95.0933	nan	95.0933
6	iDriveVLA	94.9499	nan	94.9499
7	TOAD	94.8798	nan	94.8798
8	ChainFlow-VLA	94.8468	nan	94.8468
9	DriveR	94.5945	nan	94.5945
10	vault	94.5817	nan	94.5817
11	CLOVER	94.4867	nan	94.4867
12	yinbuaa	94.2075	nan	94.2075
13	Lead2026	94.1726	nan	94.1726
14	AMC-Drive	93.9444	nan	93.9444
15	RAP	93.798	nan	93.798

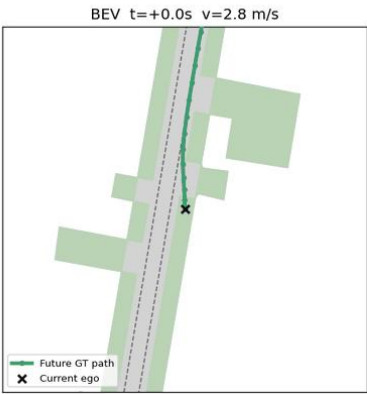
KEY RESULTS

NAVSIM Benchmark Example

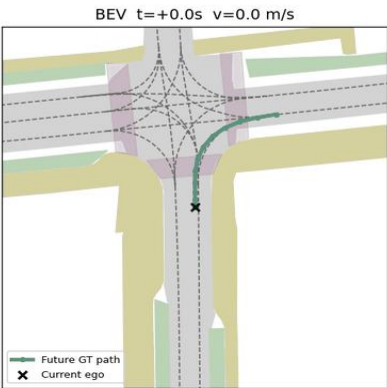
NAVSIM v1 driving process | 026 | a7d3076a7b7b58b2 | frame 1/14 | 2021.10.06.07.26.10_veh-52_00772_00917



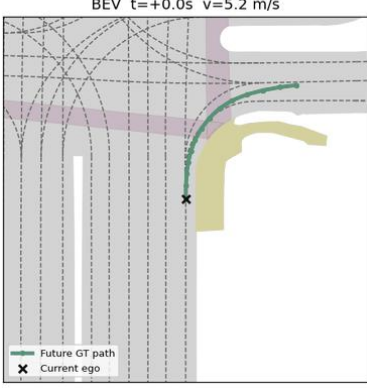
NAVSIM v1 driving process | 041 | c678ab21b707538f | frame 1/14 | 2021.09.16.17.40.09_veh-45_02539_02745



NAVSIM v1 driving process | 018 | bf1d8c493e7458d2 | frame 1/14 | 2021.08.30.13.45.25_veh-40_00878_01104



NAVSIM v1 driving process | 053 | cc853dd8697f51b3 | frame 1/14 | 2021.05.25.14.16.10_veh-35_04097_04328

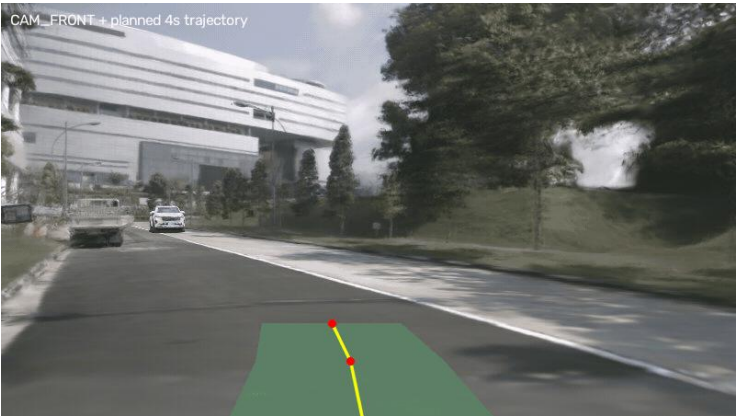
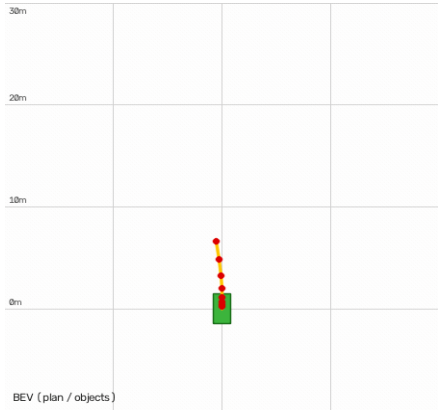
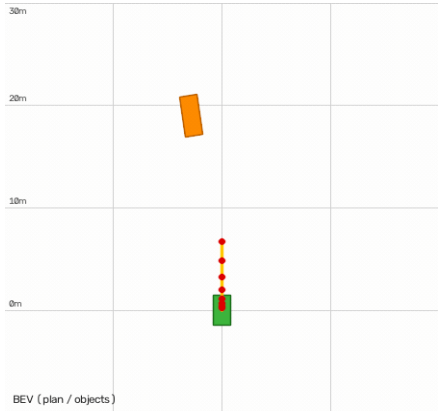


KEY RESULTS

Closed-loop Evaluation on HUGSIM

HUGSIM 0.43

	Metric	Baseline (Ours) [†]	WA-JEPA	LTF	DrivoR	UniAD	VAD
All scenarios	NC	0.7258	0.6856	0.4428	0.5217	0.6555	0.4117
	DAC	0.9578	0.9635	0.9275	0.9559	0.9320	0.9028
	TTC	0.6672	0.6120	0.3751	0.4620	0.5156	0.2798
	Comf.	0.8567	0.6620	0.9478	0.9390	0.6633	0.9534
	PDMS	0.6374	0.5717	0.3653	0.4475	0.4940	0.2831
HD-Score by difficulty	RC	0.5312	0.5689	0.3804	0.4721	0.4383	0.3006
	HD-Score	0.4308	0.4462	0.2310	0.3252	0.3124	0.1393
	Easy (n=80)	0.7777	0.7977	0.6608	0.7799	0.6395	0.4197
	Medium (n=149)	0.5387	0.5563	0.1547	0.2911	0.3718	0.0849
	Hard (n=94)	0.2629	0.3060	0.1204	0.2000	0.2099	0.0770
	Extreme (n=99)	0.1476	0.1362	0.1167	0.1407	0.0632	0.0626

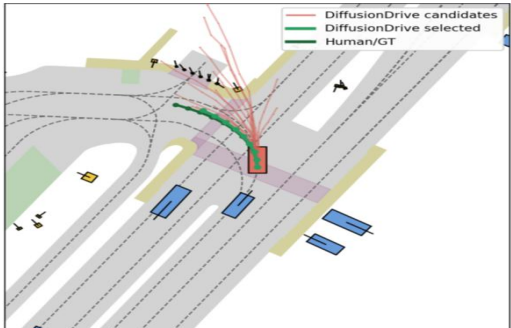


KEY RESULTS

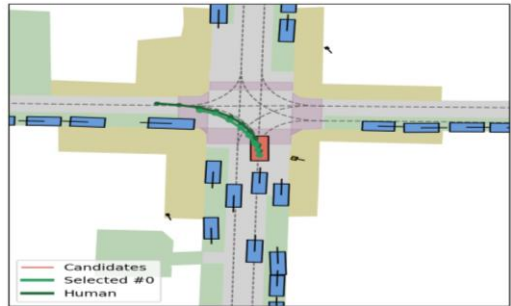
De-redundant Multimodal Planning Framework



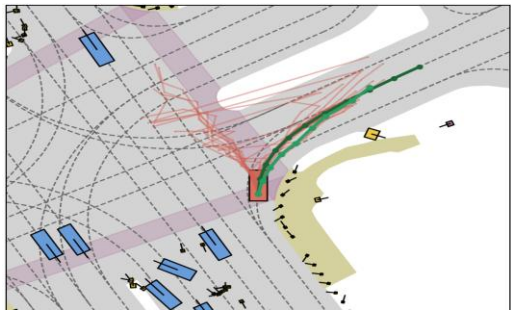
Diffusion Drive (Diffusion Model)



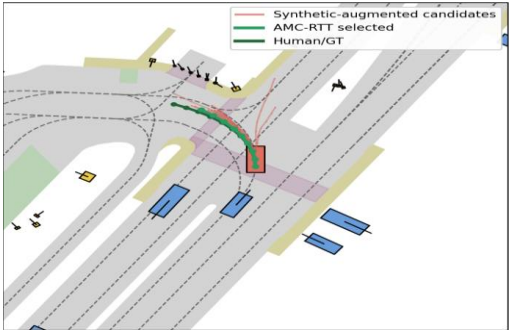
RecogDrive (FlowMatching)



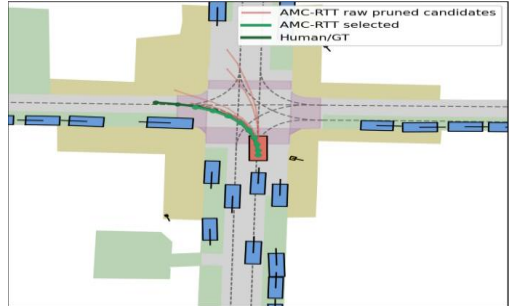
Drive-jepa (Proposal_based)



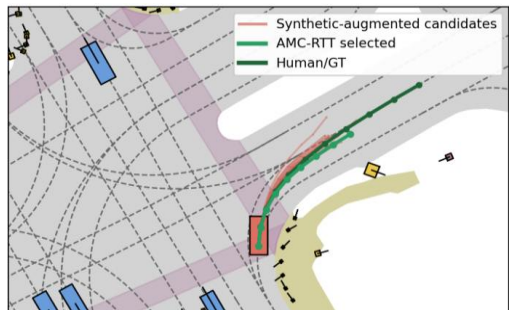
Diffusion Drive_AMC



RecogDrive_AMC



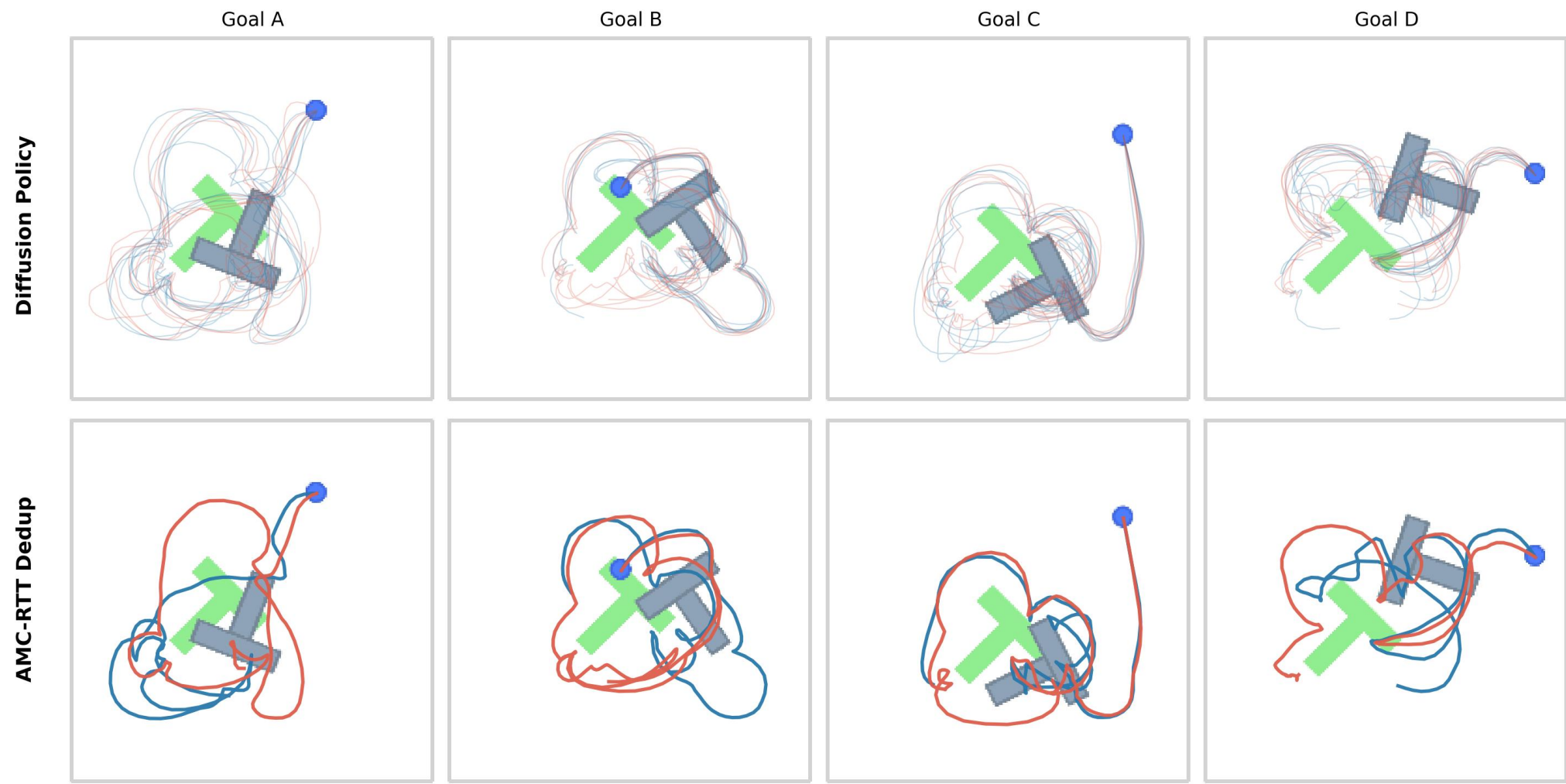
AMC-Drive



[8] C. Chi et al., "Diffusion Policy: Visuomotor Policy Learning via Action Diffusion," RSS, 2023.

KEY RESULTS

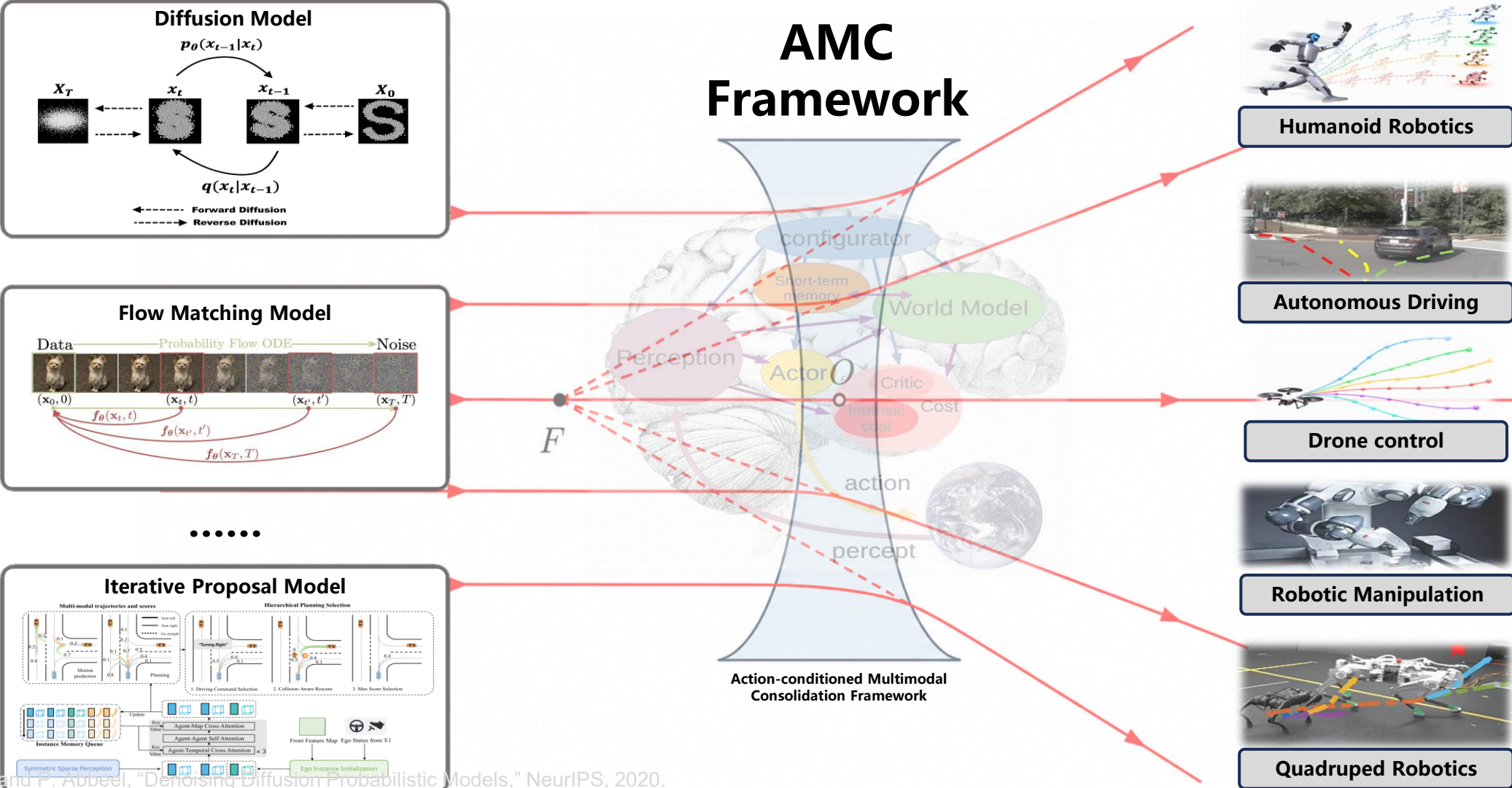
De-redundant Multimodal Planning Framework



[8] C. Chi et al., “Diffusion Policy: Visuomotor Policy Learning via Action Diffusion,” RSS, 2023.

KEY RESULTS

De-redundant Multimodal Planning Framework



[8] J. Ho, A. Jain, and P. Abbeel, "Denoising Diffusion Probabilistic Models," NeurIPS, 2020.
[9] Y. Lipman et al., "Flow Matching for Generative Modeling," ICLR, 2023.
[10] W. Sun et al., "SparseDrive: End-to-End Autonomous Driving via Sparse Scene Representation," ICRA, 2025.